

National Institute of Business Management
School of Computing and Engineering
Course work | Assessment Announcement Sheet



Course Name	HDSE HNDIS		
Module Name	Data Management 2		
Batch			
Learning Outcomes Covered (Mention according to the Module Descriptor)	1. Identify the fundamentals of Oracle Database Server / SQL Server		
	2. Explain PL/SQL programming Techniques.		
	3. Apply Oracle for Database administration		
	4. Use Non-Relational Databases		
Assessment CW No	1		
Assessment Mode	Group	Group (if it is group mode only)	
		Group Size	Grouping Criteria
		4	Maximum number of members per group is 4
Assessment Type	Software- Presentation VIVA		
	If other specify		
Hand in Date Time	31.08.2026		
Hand out Date Time	28.09.2026		
Submission Details (Format and Location)	GitHub link of the project		
Plagiarism Acceptance Level	AI generated contents will obtain 0 marks.		

Assessment | CW Description

Note:

This is a take home assessment

Description:

You are a database developer working for a transportation service provider called **SmartMove Transport Solutions**. The company provides passenger transportation services through buses, vans, and other vehicles for daily commuters, corporate employees, and special travel requirements.

The system manages vehicle operations, driver assignments, passenger registrations, route planning, trip scheduling, ticket reservations, payment processing, and customer feedback. The platform aims to improve transportation efficiency by providing real-time trip management, digital booking facilities, and centralized transport operations.

The system offers a wide range of functionalities, including vehicle management, route management, driver management, passenger management, online ticket booking, payment processing, and trip monitoring.

- Design the ER or EER diagram.
- Develop a simple Web or Enterprise Application for the SmartMove Transport Solutions system using any suitable programming language or framework.
- Design and implement a relational database schema using Oracle Database.
- The database must support core system operations, including:
 - Vehicle Management
 - Driver Management
 - Route Management
 - Passenger Management
 - Trip Scheduling Management
 - Ticket Booking Management
 - Payment Processing
 - Maintenance Management
 - Feedback and Review Management
- Populate the database with sufficient sample data to simulate a realistic application scenario.
- Write and implement PL/SQL programs (procedures, functions, cursors, and/or triggers with exception handling) to perform database operations and business logic.
- Your PL/SQL implementation should include at least five(05) business reports.

Examples

- Identifying the most frequently used routes based on ticket bookings.
- Calculating total revenue generated from ticket sales within a given period.
- Generating passenger travel history reports.
- Finding vehicles due for maintenance.

MongoDB Integration

- Implement MongoDB as a complementary NoSQL database to manage unstructured and dynamic transportation-related information.
- MongoDB collections should capture:
 - Vehicle images and documents.

- Passenger reviews and feedback comments.
- Travel announcements and notifications.
- Trip-related multimedia content.
- Ensure data is stored in a schema-flexible format to support dynamic content and community interactions.
- Demonstrate queries in MongoDB to,
 - Retrieve all passenger reviews for a specific route.
 - Identify highest-rated vehicles or drivers based on customer feedback.
 - Search customer complaints or feedback using specific keywords.
 - Retrieve vehicle documents and multimedia information.
- Integrate MongoDB with the application layer to provide real-time content delivery and interactive forums for students and instructors.

Deliverables:

- Presentation
- Software Application with Database Implementation

Marking Rubric

Criteria	Weight	0% – 39%	40% – 49%	50% – 59%	70%+
Front-end Development + ER Diagram	20%	Very poor or missing front-end, ER diagram incomplete or incorrect	Basic front-end with minimal functionality, ER diagram partially correct	Functional front-end covering most operations, ER diagram mostly accurate	Fully functional, user-friendly front-end, ER diagram complete, normalized, and clearly represents all entities
Database Implementation (Oracle)	15%	Incomplete DB implementation, fails to meet requirements, no PL/SQL usage	Few operations (1–2) implemented with PL/SQL, limited security mechanisms	Two-thirds of operations implemented, some security mechanisms, acceptable backup plan	All operations implemented with PL/SQL, correct schema, strong security and backup mechanisms
MongoDB Incorporation	15%	No MongoDB implementation	Minimal MongoDB requirements captured, weak queries	MongoDB implemented with requirements, some queries demonstrated, partial integration	Correct MongoDB incorporation, requirements fully captured with standard queries
Reports & Business Logic (PL/SQL)	15%	No meaningful reports, poor exception handling	Few reports generated, limited exception handling	Reports implemented, acceptable exception handling	Comprehensive reports, robust PL/SQL logic, advanced exception handling
Completeness of the Project	5%	Incomplete, major components missing	Less than half completed	More than half completed but missing some features	Fully complete, all requirements achieved

Presentation & Viva	20%	Poor understanding; weak communication	Average understanding; average communication	Good understanding, clear communication	Excellent understanding, confident communication, demonstrates deep knowledge
Integration & Innovation	10%	No integration between Oracle and MongoDB	Minimal integration, weak linkage between systems	Partial integration, some cross-system use cases	Strong integration, innovative use of Oracle + MongoDB for hybrid analytics